

Goal. Get back into the swing of calculus with some integrals!

1. Evaluate the integral by direct antidifferentiation:

(a) $\int (2x + 3x^{4/5}) \, dx$

(b) $\int 2e^x \, dx$

(c) $\int \frac{10}{x} \, dx$

(d) $\int \sec^2(x) \, dx$

(e) $\int \frac{4}{\sqrt{1-x^2}} \, dx$

(Hint: What are the derivatives of the inverse trig functions?)

2. Use substitution to evaluate the following integrals. Write down the substitution you use.

(a) $\int 4 \cos(7x) \, dx$

(b) $\int \frac{5}{x \ln x} \, dx$

(c) $\int_0^1 x^2 \sqrt{4x^3 + 5} \, dx$

(d) $\int \frac{\cos(3x)}{e^{\sin(3x)}} \, dx$

(e) $\int_0^x u^3 \sqrt{1 + u^2} \, du$

3. What do you need to remember when doing u -substitution?
4. How do you decide which u -substitutions to try?
5. Write an integral whose solution involves u -substitution. Solve this integral.